Evolving a Population Code for Multimodal Concept Learning

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***Abstract*—We describe an evolutionary method for learning concepts of objects from multimodal data. The proposed method uses a population code (hypernetwork representation), i.e. a collection of codewords (hyperedges) and associated weights, which is adapted by evolutionary computation based on observations of positive and negative examples. The goal of evolution is to find the best compositions and weights of hyperedges to estimate the underlying distribution of the target concepts. We discuss the relationship of this method with estimation of distribution algorithms (EDAs), classifier systems, and ensemble learning methods. We evaluate the method on a suite of image/text benchmarks. The experimental results demonstrate that the evolutionary process successfully discovers salient codewords representing multi-modal feature combinations for describing and distinguishing different concepts. We also analyze how the complexity of the population code evolves as learning proceeds.**

***Index Terms*—Population based evolution, distribution approximation, ensemble learning**

I. Introduction

Learning a concept in a multimodal (image/text) environment is important for many applications in artificial intelligence and cognitive science, including computer vision, natural language processing and robotics. Inflexible learning methods with rigid model structure are not suitable for learning multimodal concepts. In this paper, we introduce an evolutionary method that learns a population code to represent multimodal concepts. The proposed method approximates the underlying distribution of features by evolutionary computation. It represents a concept collectively using a set of randomly sampled feature combinations. Through evolution, new concepts emerge by combining existing feature combinations and a proper weight is assigned to each feature combination representing its contribution to the concept.

The proposed method consists of the following steps: First, a set of hyperedges (randomly sampled features or codewords) constitute an initial population. By changing the weight to each hyperedge and generating new hyperedges, the method evolves a population code consisting of $<H, W> = \{(h_i, w_i)|h_i$: each hyperedge, w_i: associated weight$\}$. Variational operators change the composition and distribution of hyperedges. From machine learning point of view, this is an evolutionary learning method which builds the best concept descriptor by approximating a underlying distribution of features.

We note that i) the proposed method evolves a population of a large number of weak learners, ii) it approximates the underlying distribution of a target concept without assuming a prior distribution, and iii) it finds important building blocks and adapts them through evolution. By employing the idea of weak learning and feature distribution approximation, we introduce a very flexible representation which is a population code.

The proposed method is used for image classification tasks in a multimodal (image/text) environment. Many images come with text data such as captions or related articles. If we associate more relevant textual words to each image category, it is possible to improve the image classification in the given article although the image is untagged. We use an autonomous learning method where multimodal hyperedges (visual words + textual words) are generated and a proper hyperedge distribution for each image category is approximated in an evolutionary manner. Because it is difficult to determine an optimal set of hyperedges for each image category in advance, we generate hyperedges of low orders and then combine them using variation operators. By generating hyperedges and adjusting weights at the same time, the population $<H, W>$ is evolved.

We evaluate the proposed method on image datasets including PASCAL 2008 images [2] and Caltech 101 [3]. We report on our findings, which are very promising both from machine learning and evolutionary computation points of view. Hyperedges of relatively lower orders indicate that the proposed method is able to discover general properties which are not very specifically restricted to a small number of instances. We observe that the evolutionary learning process can find building blocks and preserve them. We conclude that the proposed method evolves a suitable solution for the given task.

This paper is organized as follows. In Section II, we review the related works. Section 3 explains the proposed learning model. Section 4 presents experimental results. In Section 5, we discuss the characteristics of the proposed method and summarize this work.

II. Related Works

Here we review some previous researches related with our work. A learning classifier system (LCS) consists of a set of rules to model an intelligent decision maker collectively [10]. The motivation of LCS is that, when dealing with complex systems, seeking a single best-fit model is less desirable than evolving a population of rules which collectively model the system. LCS consists of four components: a finite population of classifiers, a performance component, a reinforcement component, and a discovery component. Our method is also motivated to seek a method where a large set of feature combinations represents a solution. However, our method approximates the underlying distribution by repeating the sampling and generation procedure. Our method introduces the essential diversity by changing the distribution. In terms of density estimation, EDA (Estimation of Distribution Algorithm) is also related [12], [13]. In EDA, the representation of population is replaced with a probability distribution over the available choices. Our method also tries to approximate a distribution. In our approach, however, a population is not a set of promising individuals. A population itself is a solution and each code corresponds to a feature combination. Though in a different context, recently a similar idea of EDAs based on Markov networks was presented in [9], where the Markov network can be viewed as a special case of the hypernetwork [4] if the hyperedges are constructed by cliques only.

Diversity is essential in evolutionary computation. In order to avoid dominance of a single individual, various methods have been suggested such as Sequential Niche Technique [14], dual population GA [7] or multi-population GA (MPGA) [6], [15]. MPGA maintains diversity by extra population pools. Although employing various approaches, the problem of dominance of a single individual is inherent. Our method approximates a distribution by sampling and generation, avoiding this problem.

The idea of using a population of a set of weak learners has been successfully used for machine learning [1]. Each hyperedge in our representation is a kind of weak learners. Each codeword in the population can be interpreted as weak learners with a subset of inputs. The ensemble of weak learners is learned by evolution operators. Our variation operator based on sampling and generation continuously changes the distribution.

Because keyword-based search is very expensive and incomplete, content-based image retrieval has become popular. It is reported that integrating cross-media text and visual content representation can improve the performance [16]. Our method builds a population of hyperedges. Because of its weighted hypergraph structure, our method can represent combinations of text and visual components [4]. In addition, the order (cardinality) of hyperedge is not fixed. As a result, a very flexible representation is possible.

Many digital images come with text data such as captions or articles. In order to reflect this fact, we try to associate text features to each image category. The possibility of associating words with image region is also exploited in [17], [18]. In contrast to these approaches, our method does not attempt to retrieve specific images. The proposed method attempts to improve the classification performance by associating word features to each image category.

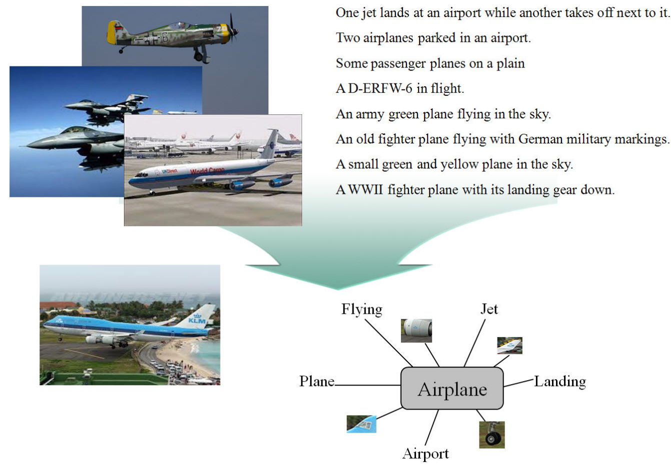

Fig. 1. Image classification in a multimodal environment. The aim of image classification is to classify a given image with text data. The proposed method evolves a population code explaining given categories. The evolved codes determines the unknown image using associated features.

We use cross-entropy for variation of hyperedges. Generating a solution of a population codewords based on cross-entropy is similar to the idea of building a global predictive model with local patterns [8]. In [8], each local pattern is viewed as a constraint on an a high-order joint distribution of interest. The cross-entropy is used to select a specific distribution. Contrary to this, we generate new feature combinations from existing feature combinations and then update the distribution.

III. Proposed Method

A. Task and Materials

We apply the proposed method to image classification task. Fig. 1 shows the task of image classification in a multimodal environment. The image classification task is to classify a category of the given image. In this task, we assume a very realistic situation where an image comes with captions or in an article and image itself is untagged. Our aim is to construct a population of code consisting of features (visual features and textual features) which are specific to a given category. Our hypothesis is that we can improve the image classification performance by incorporating textual information.

We use sentences from a dataset built by Farhadi et al [2] and images from the Caltech 101 [3]. Because we do not aim to generate a caption capable of explaining a given image, we substitute images in [2] with images in Caltech 101. Our aim is to select suitable visual words and textual words from these datasets and construct a population of hyperedges using these words (visual and textual).

TABLE I
THE SIZE OF DATA

	Image	Text
Training	300	300
Test	300	300
Number of categories	10	10

Because there are vast variations in images belonging to a same category, preprocessing of image data is inevitable. The goal of image data preprocessing is verifying and constructing visual words. SIFT (scale-invarinat feature transform) and *k*-means are used for preprocessing. SIFT [11] is used for transforming image into vector representations. There are two functions in SIFT which are Detector and Descriptor. Detector detects interesting points called key-points where property changes suddenly, such as edges. Once those interesting points are found, Descriptor is used for extracting invariant feature which consist of 128 dimension vectors. However, each image contains different number of interesting points from each other. On the average, we extract 600 SIFT features per image. For unified cardinality of representation of images, we apply *k*-means algorithm to SIFT features with *k* = 1000. This *k* value is carefully choosen after tests for the performance. Neither smaller value nor larger value is appropriate because larger *k* results sparse representation of data and smaller *k* merges different visual features as single feature. Each image is than transformed to histogram vector with dimension indicatingcentroids derived from *k*-means algorithm and height representing the number of corresponding cenroids in that image. For easier computation each histogram is actually transformed to boolean representation.

We formulate text data concerning each image as bag of words. The library for the vector representation of text data is preprocessed. The only top 1000 most frequent words are used as library. For the multi-modal experiment we just adjoin image data array with text data array. For text, we remove stop words such as “the”, “an” in order to improve the quality of hyperedges.

B. Method

In the proposed method, we regard the whole population as a solution and each hyperedge as a individual feature. The role of variation operator is to generate new individual by changing the population distribution. Through evolution, novel hyperedges are emerged. But these new hyperedges do not represent candidate solutions for the given problem. They represent features constituting an individual and contribution of each hyperedge is determined through its weight. The proposed method has following advantages. Firstly, the individual structure is very flexible. At the initial step, there exists only basic features (low order hyperedges). However, novel features (higher order hyperedges) emerge through evolution using the proposed operator in Algorithm 1. Because we do not impose any restriction on individual structure such as length or possible combination of features, it is possible to generate a solution not restricted by initial population or individual structure. Secondly, the representation is very flexible. An evolved individual is represented as a set of weighted hyperedges. By adjusting the selection of hyperedges, it is possible to control the representation power of each individual. Thirdly, important feature block can be preserved. Because hyperedges with higher weights are retained though evolution, there is no danger to destruct useful building block in the course of evolution.

$$N = \sum_i I(x_c),\ \ I(x_c) = 1 \text{ for correct classification} \qquad (1)$$

Algorithm 1 Population code evolution

Input : $D = \{C_1, C_2, \cdots, C_{cl}\}$
(D: training data, cl: number of classes)
$C_j = \{\mathbf{x}_{j_1}, \mathbf{x}_{j_2}, \cdots, \mathbf{x}_{jp_j}\}$
p_j : (number of training data of the *j*th class)
$\mathbf{x}_{jk} = (x_{jk_1}, x_{jk_2}, \cdots, x_{jk_d})$
$x_{jk_i}(1 <= i \quad 1 <= d)$ is either 0 or 1
Output : $H =< V, E, W >$
Goal : $H_{strong} = \arg\max_h P$
Begin
$Initialization$: Draw N initial Hyperdeges (weak learners) according to the initial probability distribution P_0 (overlapping of hyperedges is not allowed).
while !Termination condition **do**
Evaluation: evaluate H on a fitness function F
Discarding: Discard Hyperdeges with lower F value
Generation: $h_{new} = \text{com}(h_{parent_1}, h_{parent_2})$
Selection probability: $p_{h_{\beta\alpha}}(i) = \frac{N_{h_{\beta\alpha}}(i)}{\sum_{i=1}^{cl} N_{h_{\beta\alpha}}(i)}$
end while

$$N_{h_{\beta\alpha}}(i) = \sum_{\mathbf{x}_{i_k} \in C_i} I(\mathbf{x}_{i_k}, h_{\beta\alpha}) \qquad (2)$$

where,

$$I(\mathbf{x}_{jk}, h_{\beta\alpha}) = \begin{cases} 1 & \text{if } x_{jkv_{\beta\alpha l}} = 1 \text{ for } \forall v_{\beta\alpha l} \in h_{\beta\alpha} \\ 0 & \text{otherwise.} \end{cases}$$

$$w_{h_{\beta\alpha}} = \frac{1}{CE(P_{h_{\beta\alpha}}, \delta_\beta)} = \frac{1}{\sum_{i=1}^{c_l} p_{h_{\beta\alpha}}(i) log \frac{P_{h_{\beta\alpha}}(i)}{\delta_\beta(i)}} \qquad (3)$$

Variational Operatror *com*:

$$\text{com}(h_1, h_2) = h_1 \text{ U } h_2 \qquad (4)$$

$h_{\beta\alpha} = \alpha$'th hyperdege with class label β
$\delta_\beta = cl$ dimensional vector with βth element is 1, other elements are 0

δ is an ideal classifier. This algorithm search through combinatorial space of hyperedges to find best hyperedge distribution of which is similar to delta

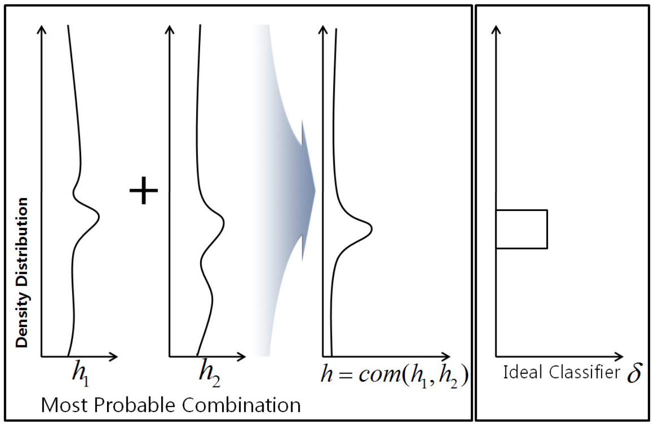

Fig. 2. Boosting weak classifiers.

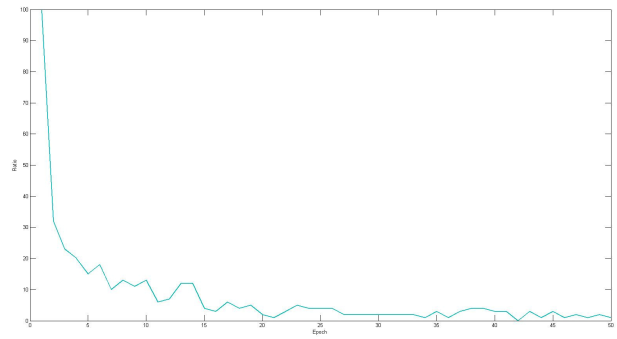

Fig. 4. Ratio of newly created Hyperedges.

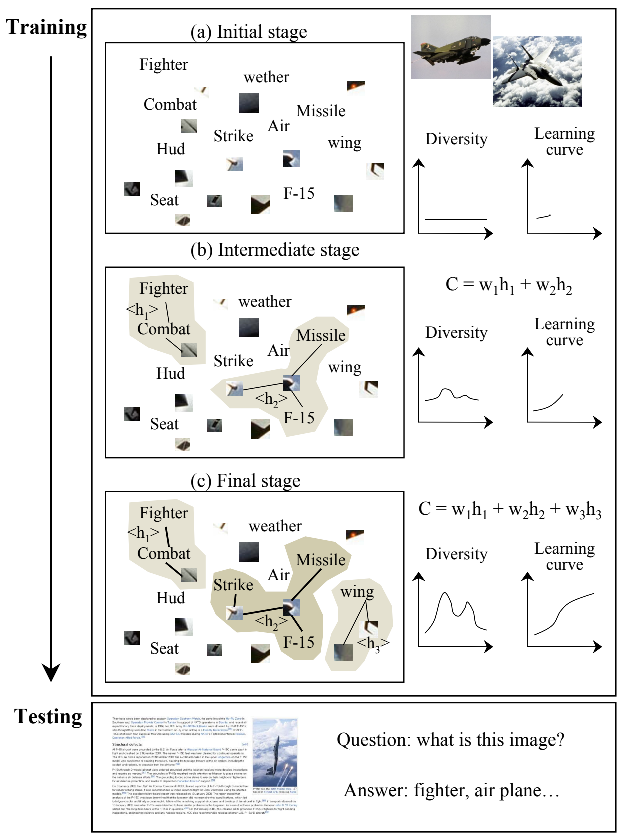

Fig. 3. Through learning, new hyperedges (feature combinations) are emerging. A concept is explained by a weighted sum of dominant hyperedges.

Two kinds of fitness functions are used. The first one is a fitness function at the hyperedge level. This fitness function is to select candidate hyperedges for the variation operator. The second one is used to verify the classification performance of an individual.

The representation scheme is based on the hypernetwork learning [4]. A hypernetwork H is defined as $H = (V, E, W)$ where V, E, and W are a set of vertices, hyperedges, and weights respectively. A hyperedge of order (cardinality) k is referred to a k-hyperedge. Fig. 3 shows the proposed evolution framework. The initial population is composed of low order hyperedges with pre-calculated weights according to the data distribution (Fig. 3, (a)). Through evolution new hyperedges of higher order are generated by joining the existing hyperedges and the weights of newly created hyperedges are calculated after every iteration (Fig. 3, (b)). As a result, the diversity of population increases. The ratio of each modality or order of each hyperedge is not restricted. The proposed method search for best concept descriptor by generating new hyperedges (feature combinations). At the end of evolution, prominent hyperedge (hyperedge (h_2)) and less prominent hyperedges (h_1, h_3) are generated. When a new image is given with textual information, the proposed method can identify the query using the image information and textual information at the same time by using the obtained hyperedges. Each class model has it's own hypernetwork models. When a new query comes in, the algorhthm adds all the weights of corresponding hyperedges in the class. The class with the highest weight sum is assigned to the new data.

IV. Experimental Results

A. Emergence of New Features

In the proposed method, the role of variation operator is to introduce diversity by changing the distribution. Fig. 4 shows what happens during evolution. Fig. 4 depicts the ratio of newly created hyperedges at each epoch. At the early stage, the population is composed of heterogenous hyperedges. However, the ratio of newly created hyperedges is rapidly declined and the composition of the population is stabilized after epoch 20. This behavior is in line with image classification results in Fig. 8, 9, and 10. The stabilization can be explained as follows. Within relatively few iterations, hyperedges representing important patterns obtain enough weights. Because we adopt entropy operation for our variation operator, these edges dominates the population.

Fig. 5 shows the trends in hyperedge order. At the 0th epoch, there exists only hyperedge of order 1. However, the ratio of order 2 and order 3 increases rapidly and the ratio of order

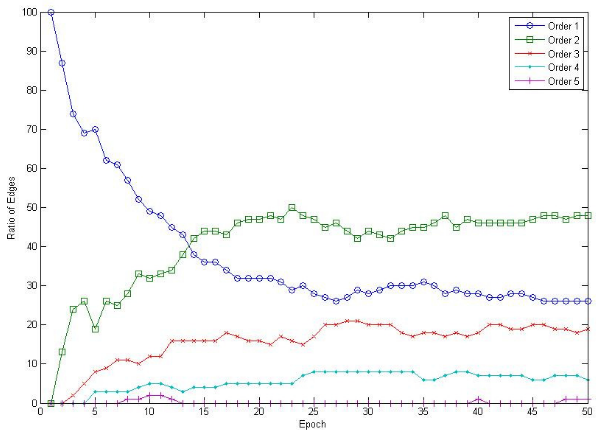

Fig. 5. Evolution of the number of hyperedge orders.

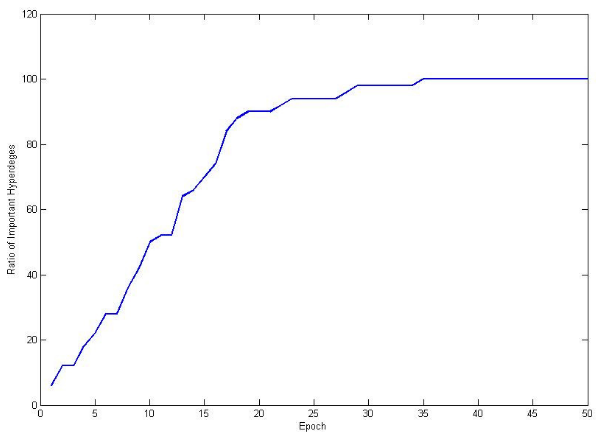

Fig. 6. Ratio of important building blocks in each epoch.

1 decreases. The dominance of lower order hyperedge shows that the classifiers for the given categories can be explained without too specific hyperdges.

B. Hyperedges as Schemas

Instead of one individual for each solution, our method regards a set of hyperedges as a solution. Because an individual is a weakly combined set of hyperedges, there is an important benefit in the proposed method. When a hyperege with critical contribution is emerged, the hyperedge can be preserved. This is due to the fact that the proposed method creates new hyperedge by combining existing ones and adjusts the contribution of each hyperedge using weight of each hyperedge. In this scheme, the preserved hyperedges can be treated as important building blocks (schemas [5]) free from danger of destruction by variation operator.

Fig. 6 shows the rate of preserved hyperedges. The ratio in Fig. 6 is calculated as the ratio of important edges constituting the final solution at each epoch. Fig. 6 shows a similar trend in Fig. 8, 9, and 10. After the ratio is above 50%, the image classification performance is stabilized. Because our method introduces the diversity by changing the distribution of hyperedges, we can find a satisfiable solution while preserving building blocks of each solution.

Stability in Fig. 6 may imply lack of exploration compared to the conventional evolutionary computation methodology. This question can be answered by another experiment with much larger pool of initial population. However, we leave this experiment for future works.

C. Evolved Population Code

The proposed method evolves a set of hyperedges describing concepts through evolution. A set of hyperedges having general properties is needed to describe a concept successfully. Fig. 7 shows the evolved hyperedges for categories “bird”, “potted plant”, and “airplane.” Because the sentences in [2] are used to explain given images, those sentences are very heterogeneous. For example, sentences such as ”A D-ERFW-6 in flight”, “An army green plane flying in the sky”, “An old fighter plane flying with German military markings”, and “A WWII fighter plane with its landing gear down” are used to describe Focke-Wulf 190. For the concept of *airplane*, the final hyperedges contain words such as “airplane”, “white”, “passenger”, and “runway.” For the concept of *bird*, the final hyperedges contain words such as “bird”, “water”, “perched”, and “beak.” These words are very general to each category. For other categories, the evolved hyperedges are also composed of general words despite of heterogeneous sentences. These generality explains the good image classification performance in Fig. 10.

D. Image Classification

In this section, we report image classification results. Fig. 8 shows image classification result when each hyperedge is composed of only visual words. Fig. 9 is obtained when each hyperedge is made from textual words. Therefore, this figure shows the result of text classification rather image classification. For hyperedges composed of visual and textual words, Fig. 10 reports the image classification performance. Although we preprocessed the image data, it is difficult to obtain useful visual words representing a category. Therefore, the test performance in Fig. 8 is very bad. Regarding the heterogeneity in text data, classification result based on textual words is relatively good. Fig. 10 reports image classification results when both of visual words and textual words are used to generate hyperedges. It is apparent that hyperedges made of multimodal data contribute greatly for image classification when compared Fig. 8 and Fig. 10. Fig. 8 and Fig. 10 show that our initial hypothesis is useful. By incorporating additional text information, we show that it is possible to improve the image classification performance.

V. Discussion and Conclusion

In this paper, we introduced an autonomous concept learning method capable of approximating feature distributions by evolutionary computation. Our contributions are as follows.

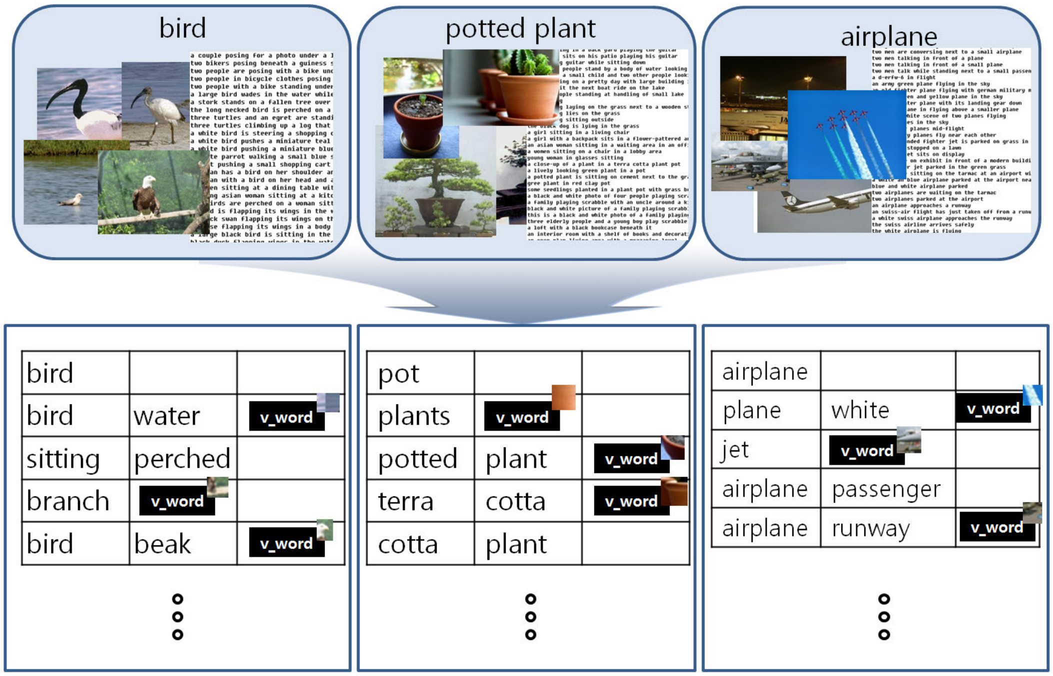

Fig. 7. Dominant hyperedges for categories "bird", "potted plant", "airplane." "*v_word*" means "*visual word*." In spite of heterogenous sentences, the evolved hyperedges contain plausible words for each category, such as "jet" for airplane and "branch" for bird.

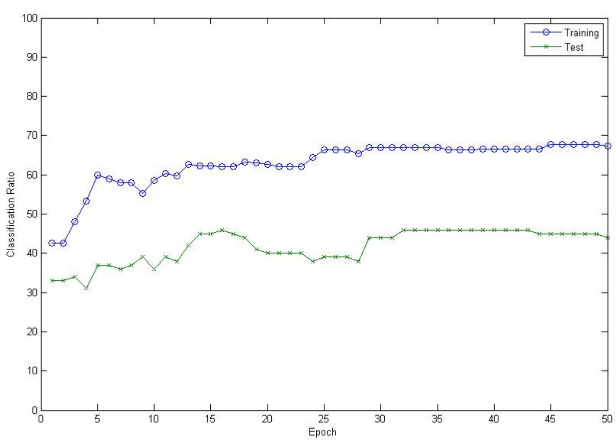

Fig. 8. Classification results using only visual words.

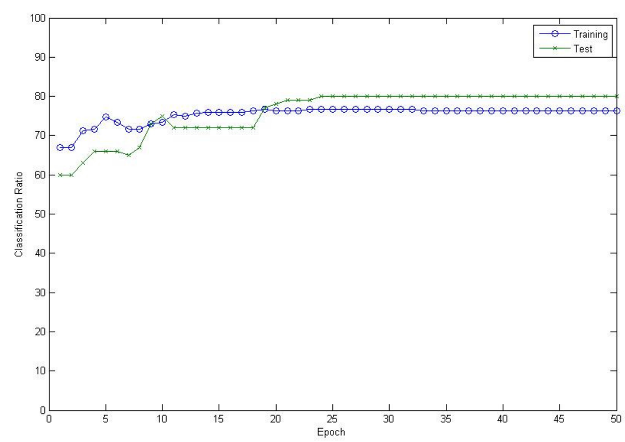

Fig. 9. Classification results using only text words.

First, our method is a promising evolutionary method based on the idea of population code. Because our method builds new hyperedges through combining existing hyperedges, our method is not restricted by the initial population and the individual structure. Second, our method allows flexible description for concepts. Because our method regards each hyperedge as a weak learner and collectively builds a concept descriptor based on the contribution of each hyperedge, it is possible to represent each concept in a very flexible manner. Third, our method can identify prominent feature combinations more easily and maintain them more effectively than conventional evolutionary computation. Because our method assign more weights to more contributing hyperedges and introduces diversity by changing the distribution of features, it is possible that the prominent hyperedges can maintain their characteristics through entire evolution process. Therefore, we introduced an interesting framework capable of preserving schemas in an individual very easily. Forth, we introduce an interesting method capable of processing multi-modal data. Because the proposed method searches for prominent visual word - textual word combinations in dynamic manner, it is possible to constitute useful multimodal descriptors empirically.

Although the proposed method introduces interesting characteristics, there exist various aspects where further discussion

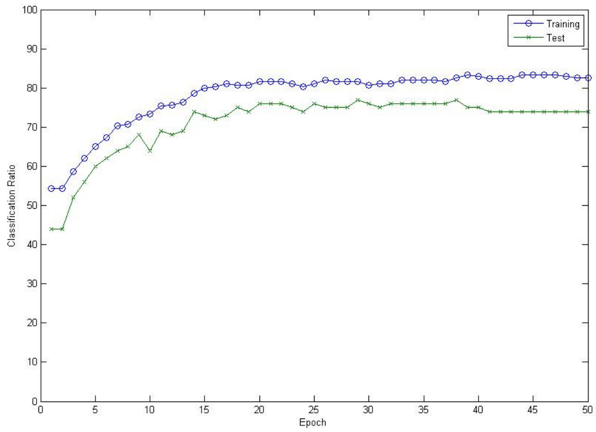

Fig. 10. Classification results using both of visual words and textual words.

is needed. We need to consider the selection pressure. The proposed method approximates the underlying distribution by repeating sampling and generation. Entropy is used to control this sampling and generation procedure. Although entropy is very useful, we can use other methods to control the selection pressure. Because it is possible that the underlying distribution is a multinomial one, we can assume the Dirichlet distribution and employing other methods such as Gibbs sampling and resampling concept for approximation [19].

We also need to consider the computational advantages. From Fig. 5, we can assume that a population of hyperedges with relatively lower order is sufficient for representing a solution. If a difficult task is given, a typical genetic algorithm would require very complex individual structure. Therefore, the computational cost for evaluating each individual would increase. Contrary to this computational cost, lower order hyperedges is likely to require less computational burden. If we are able to restrict the size of population needed to estimate the underlying distribution, we can lessen the computational burden.

Classification performance in Fig. 9 and Fig. 10 needs more explanation. With the given performance, it is easy to assume that textual information is sufficient for the given task. However, this assumption is very short-sighted. In a multimodal environment where dynamic interaction is a norm, one needs a novel data structure capable of processing vast interaction. Because hyperedges are able to represent multimodal information in a very flexible manner, a population of hyperedges is suitable for a multimodal environment. Comparing Fig. 8 and Fig. 9 on one hand and Fig. 10 on the other, we observe that the test performance improves more smoothly and faster when the visual and textual words are used together instead of each of them alone. However, in these experiments we have used raw image patches (which is very noisy) rather than visual objects, and thus statistically more significant experiments remain to be done in the future.

big potential in Fig. 9 and Fig. 10. In this work, the contribution of visual words is insignificant because we utilized a raw visual words not image objects. We will investigate the performance improvement using image objects in further work.

Because our method does not assume prior distributions and does not estimate the parameters of underlying distributions, it has a big potential for dynamic learning. But the proposed method lacks the life-long learning categorization ability [20]. In future works, we will observe the behavior of the proposed method in dynamic learning environment such as video stream learning by enhancing the dynamic learning ability.

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REFERENCES

[1] Y. Freund, Boosting a weak learning algorithm by majority, *Information and Computation*, 121 (2), pp. 256-285, 1995.
[2] A. Farhadi, M. Hejrati, M. A. Sadeghi, P. Young, C. Rashtchian, J. Hockenmaier, and D. Forsyth, Every picture tells a story: generating sentences from images, *Proc. ECCV 2010*, pp. 15-29, 2010.
[3] http://www.vision.caltech.edu/Image_Datasets/Caltech101/
[4] B.-T. Zhang, Hypernetworks: A molecular evolutionary architecture for cognitive learning and memory, *IEEE Computational Intelligence Magazine*, 3 (3). pp. 49-63, 2008.
[5] J. H. Holland, Adaptaion in natural and artificail systems, MIT Press, 1992.
[6] F. F. Vega, M. Tomassini, W. F. Punch, III, and J. M. Snchez-Prez, Experimental study of multipopulation parallel genetic programming, *Proc, Eur, Conf. Genetic Program*, 2000.
[7] T. Park and K. Ryu, A dual-population genetic algorithm for diversity control, *IEEE Transactions on Evolutionary Computation*, 14 (6), 2010.
[8] H. Mannila and D. Pavlov, Prediction with local patterns using cross-entropy, *KDD-99*, 1999.
[9] A. Brownlee, J. McCall, S. Shakya, and Q. Zhang, Structure learning and optimisation in a markov network based estimation of distribution algorithm , *Exploitation of Linkage Learning in Evolutionary Algorithms*, 2010.
[10] R. Urbanowicz and J. H. Moore, Learning classifier systems: A complete introduction, review, and roadmap, *Journal of Artificial Evolution and Application*, vol.2009, 2009.
[11] D. G. Lowe, Object recognition from local scale-invariant features, *Seventh International Conference on Computer Vision (ICCV'99)*, vol.2, pp. 1150-1157, 1999.
[12] P. Larranaga and J. A. Lozano (Eds.), Estimation of distribution algorithms: A new tool for evolutionary computation. *Kluwer Academic Publishers*, Boston, 2002.
[13] Q. Zhang and H. Muhlenbein, On the convergence of a class of estimation of distribution algorithms, *IEEE Transactions on Evolutionary Computation*, 8 (2), pp. 127-136, 2004.
[14] D. Beasley, D. R. Bull, and R. R. Martin. A sequential niche technique for multimodal function optimization, *Evolutionary Computation*, 1 (2), pp. 101-125, 1993.
[15] E. Cantu-Paz, A survey of parallel genetic algorithms, vol.10, 1998.
[16] M. Rerecatu, N. Boujemaa, and M. Crucianu, Semantic interactive image retrieval combining visual and conceptual content description, *Multimedia Systems*, vol.13, pp. 309-322, 2008.
[17] Y. M. Hironobu, H. T. Takahashi, and R. Oka, Image-to-word transformation based on dividing and vector quantizing images with words, *In Boltzmann Machines, Neural Networks*, pp. 405-409, 1999.

[18] J. Jeon, V. Lavrenko, and A. R. Manmatha, Automatic image annotation and retrieval using cross-media relevance models, *Proceedings of the 26th annual international ACM SIGIR conference on Research and development in informaion retrieval*, pp. 119-126, 2003.

[19] A. Doucet, and A. M. Johansen, A tutorial on particle filtering and smoothing: Fifteen years later, 2008.

[20] S. Kirstein, H. Wersing, and E. Korner, Towards autonomous bootstrapping for life-long learning categorization tasks, *(IJCNN) The 2010 International Joint Conference on Neural Networks*, pp. 1-8, 2010